

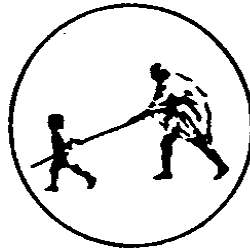
MGM Smart Parking System

BY

**Akhilesh Nagnath Phulari
Udayan Ganesh Dhaskanwar**

*Under the Guidance
of*

Ms. M. G. Shelke



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Mahatma Gandhi Mission's College of Engineering, Nanded (M.S.)

Academic Year 2025-26

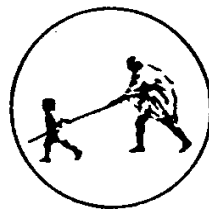
A Project Report on
MGM Smart Parking System

Submitted to
DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE

In partial fulfillment of the requirements for the completion of
THIRD YEAR BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING

Submitted By
Akhilesh Nagnath Phulari
Udayan Ganesh Dhaskanwar

Under the Guidance of
Ms. M. G. Shelke



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MAHATMA GANDHI MISSION'S COLLEGE OF ENGINEERING
NANDED (M.S.)

Academic Year 2025-26

Certificate



This is to certify that the report entitled

MGM Smart Parking System

Submitted By

Akhilesh Nagnath Phulari
Udayan Ganesh Dhaskanwar

in a satisfactory manner as a partial fulfillment of requirements for the
completion of Third Year in Computer Science and Engineering

to

MGM's College of Engineering, Nanded

Under

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE**

This work has been carried out under my guidance,

Ms. M. G. Shelke

Project Guide

Dr. A. M. Rajurkar

H.O.D

Computer Science & Engineering

Dr. G. S. Lathkar

Director

MGM's College of Engg., Nanded

ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide, **Ms. M. G. Shelke**, for the valuable guidance, continuous support, and encouragement throughout the course of this work. Working under her supervision has been a great learning experience, and we are thankful for the helpful suggestions, discussions, and motivation provided during the development of this project.

We also take this opportunity to express our heartfelt thanks to **Dr. A. M. Rajurkar**, Head of the Department of Computer Science & Engineering, for her encouragement and constant support.

We are equally thankful to **Dr. G. S. Lathkar**, Director of Mahatma Gandhi Mission's College of Engineering, for her guidance, inspiration, and valuable support.

Last but not least, we extend our sincere thanks to all those who directly or indirectly helped us in the successful completion of this project.

With Deep Reverence,

Akhilesh Nagnath Phulari[229]
Udayan Ganesh Dhaskanwar[238]
TY [CSE-B]

ABSTRACT

Parking management has become an important challenge in educational institutions due to the increasing number of vehicles and limited parking spaces. Traditional parking systems often result in traffic congestion, time wastage, manual record maintenance, and difficulties in monitoring vehicle movement. To overcome these issues, the MGM Smart Parking System has been developed as a web-based parking management solution that provides an organized and efficient approach to managing parking activities. The system is developed using HTML, CSS, JavaScript, Python, Flask Framework, and MySQL database to provide a secure and user-friendly parking management platform.

The MGM Smart Parking System includes features such as user authentication, vehicle registration, parking slot management, entry and exit monitoring, database management, and report generation. The system helps improve parking efficiency by providing accurate parking information, reducing manual work, and maintaining digital parking records. Testing and evaluation confirmed that the system performs reliably and supports efficient parking operations. Overall, the MGM Smart Parking System provides a practical, secure, and effective solution for modern parking management and contributes to improved parking organization within educational institutions.

Akhilesh Nagnath Phulari[229]

Udayan Ganesh Dhaskanwar[238]

TY[CSE-B]

TABLE OF CONTENTS

ACKNOWLEDGEMENT	I
ABSTRACT	II
TABLE OF CONTENTS	III
LIST OF FIGURES	V
LIST OF TABLES	V
Chapter 1. INTRODUCTION	1-9
1.1 Background of MGM Smart Parking System	1
1.2 Need for MGM Smart Parking System	2
1.3 What is MGM Smart Parking System?	3
1.4 Working of MGM Smart Parking System	4
1.5 Objectives of MGM Smart Parking System	4
1.6 Features of MGM Smart Parking System	6
1.7 Applications of MGM Smart Parking System	5
1.8 Technologies Used in MGM Smart Parking System	7
1.9 Future Scope of MGM Smart Parking System	8
Chapter 2. LITERATURE REVIEW	10-15
2.1 Study of Traditional Parking Systems	10
2.2 Review of Existing Digital Parking Systems	11
2.3 Research on Web-Based Parking Management Systems	12
2.4 Database Management in Parking Systems	13
2.5 Problems Identified in Existing Systems	13
2.6 Proposed MGM Smart Parking System	14
Chapter 3. SYSTEM DESIGN	16-23
3.1 Overview of System Design	17
3.2 System Architecture	18
3.2.1 Presentation Layer	19
3.2.2 Application Layer	19
3.2.3 Database Layer	20
3.3 Modules of MGM Smart Parking System	20
3.3.1 Login Module	20
3.3.2 Vehicle Registration Module	21
3.3.3 Parking Slot Management Module	21

3.3.4 Entry and Exit Monitoring Module	22
3.3.5 Database Management Module	22
3.5 Output Design	22
Chapter 4. IMPLEMENTATION	24-28
4.1 Technologies Used	24
4.1.1 Frontend Technologies	24
4.1.2 Backend Technologies	25
4.4.3 Database	25
4.2 System Implementation	26
4.3 Advantages of Implementation	27
Chapter 5. TESTING AND EVALUATION	29-36
5.1 Objectives of Testing	29
5.2 Types of Testing Performed	30
5.2.1 Unit Testing	30
5.2.2 Integration Testing	30
5.2.3 System Testing	31
5.2.4 User Interface Testing	31
5.2.5 Database Testing	31
5.3 Testing of System Modules	31
5.3.1 Login Module Testing	32
5.3.2 Vehicle Registration Module Testing	32
5.3.3 Parking Slot Management Testing	33
5.3.4 Entry and Exit Monitoring Testing	33
5.3.5 Database Management Testing	33
5.4 Evaluation of System Performance	34
5.5 Advantages Observed During Testing	34
5.6 Limitations Observed During Testing	35
CONCLUSION	37
REFERENCES	38

List of Figures

Figure No.	Name of Figure	Page No.
1.1	Technologies Used	8
2.1	Traditional Parking Problem	11
3.1	Overview of Parking System	17
3.2	System Architecture	19
4.1	Technology Stack	25
4.2	Admin Login	26
4.3	Sign Up Page	26
4.4	Parking Slots	27
4.5	Benefits of Smart Parking System	28
5.1	Testing Performance	30
5.2	User Login	32
5.3	Parking Booking Slot	32
5.4	Parking Slots	33
5.5	Database Connection	34

List of Tables

Table No.	Name of Table	Page No
2.1	Problems in Traditional Parking Systems	13

INTRODUCTION

In today's fast-growing world, transportation plays an important role in everyday life. The number of vehicles on roads is increasing continuously because people depend heavily on personal transportation for education, work, business, healthcare, and travel. Due to this rapid increase in vehicles, parking management has become one of the biggest challenges in schools, colleges, hospitals, shopping malls, railway stations, airports, offices, and smart cities. In many institutions and organizations, parking systems are still managed manually. Drivers usually spend a large amount of time searching for empty parking spaces. During busy hours, parking areas become overcrowded, leading to traffic congestion, fuel wastage, environmental pollution, confusion, and frustration among vehicle owners. Manual parking systems are unable to handle large numbers of vehicles efficiently because they depend completely on human effort and observation.

The MGM Smart Parking System is developed as an intelligent solution to overcome these parking problems. It is an automated parking management system designed to improve the process of parking vehicles in an organized and efficient way. The system helps users easily identify available parking spaces and allows administrators to monitor parking activities digitally.

The MGM Smart Parking System uses modern technologies such as web development, databases, sensors, automation concepts, and real-time monitoring to improve parking management. The system maintains vehicle records, parking slot information, entry and exit details, and parking status in a digital format. It reduces manual work and provides a secure and user-friendly parking experience. The main purpose of developing this system is to save time, reduce traffic congestion, improve parking efficiency, and provide better parking management within the MGM campus or organization. The project also supports the concept of smart infrastructure and digital transformation in modern institutions.

1.1 Background of MGM Smart Parking System

Parking management has always been an important part of transportation systems. In earlier days, the number of vehicles was less, so parking management was simple and

easy. People could easily find parking spaces without much difficulty. Parking records were maintained manually by security guards or parking staff.

However, with rapid urbanization and population growth, the number of vehicles increased significantly. Educational institutions, hospitals, offices, shopping centers, and public places started facing parking-related problems. Traditional parking methods became inefficient because they could not manage the increasing number of vehicles properly. In manual parking systems, drivers are required to move around parking areas searching for empty spaces. This consumes a lot of time and fuel. Security guards or parking staff manually monitor vehicle entry and exit, which increases the chances of human errors and poor management. Sometimes parking records are lost or maintained incorrectly.

The development of modern technologies created opportunities to improve parking management systems. Digital parking systems were introduced to automate parking activities and provide better parking control. Technologies such as sensors, RFID, web applications, databases, cameras, and IoT devices started being used for smart parking management. The MGM Smart Parking System is developed based on these modern technologies to provide a reliable, secure, and efficient parking solution. The system helps improve parking organization and supports modern smart campus management.

1.2 Need for MGM Smart Parking System

The need for the MGM Smart Parking System is increasing day by day because traditional parking systems are no longer efficient for managing large parking areas. The increasing number of vehicles creates many parking-related issues that require an intelligent and automated solution.

One of the major problems in traditional parking systems is traffic congestion. Drivers spend a lot of time searching for parking spaces, which creates unnecessary vehicle movement inside parking areas. This increases congestion and causes delays for other users. Another major problem is time wastage. In busy parking areas, drivers may spend several minutes searching for empty parking slots. This creates frustration and reduces productivity. Fuel consumption also increases because vehicles continue moving while searching for parking spaces.

Environmental pollution is another important concern. Unnecessary fuel consumption increases carbon emissions and contributes to air pollution. Smart parking systems help reduce this problem by minimizing unnecessary vehicle movement. Manual parking management also creates several operational problems. Parking records may not be maintained properly, and there is always a possibility of human error. Security monitoring becomes difficult in large parking areas without digital systems.

The MGM Smart Parking System is developed to solve all these problems by providing automated parking management, real-time parking information, accurate record maintenance, and efficient monitoring.

1.3 What is MGM Smart Parking System?

The MGM Smart Parking System is a web-based and automated parking management system that helps manage parking activities efficiently using digital technologies. The system is designed to improve parking management by reducing manual work and providing real-time information about parking availability. The system allows users to check available parking spaces before parking their vehicles. Users can register their vehicle details, and the system automatically allocates parking slots based on availability. Vehicle entry and exit details are stored in a database for monitoring and security purposes.

The system can also provide parking reports and parking statistics for administrators. It improves transparency, efficiency, and security in parking management.

The MGM Smart Parking System may include the following components:

- User login system
- Parking slot management
- Vehicle registration
- Database management
- Entry and exit monitoring
- Parking status updates
- Report generation
- Security monitoring

The system provides an organized and modern approach to parking management.

1.4 Working of MGM Smart Parking System

The MGM Smart Parking System works through several automated steps that help manage parking efficiently.

Step 1: User Registration

The user enters vehicle information such as vehicle number, owner details, and contact information into the system. This information is stored securely in the database.

Step 2: Login Process

Users and administrators log into the system using usernames and passwords. Authentication ensures secure access to parking data.

Step 3: Parking Slot Detection

The system checks the availability of parking slots using stored parking information or sensors.

Step 4: Parking Slot Allocation

An available parking slot is assigned automatically to the user.

Step 5: Vehicle Entry Monitoring

The system records the vehicle entry time and updates the parking slot status as occupied.

Step 6: Real-Time Parking Updates

Parking availability information is continuously updated in the database.

Step 7: Vehicle Exit Monitoring

When the vehicle exits the parking area, the exit time is recorded and the parking slot becomes available again.

1.5 Objectives of MGM Smart Parking System

The MGM Smart Parking System is developed with the aim of improving parking management through automation and digital technology. In many institutions and organizations, parking management becomes difficult because of the increasing number of vehicles and limited parking spaces. Traditional parking systems consume more time

and require continuous manual monitoring. To overcome these challenges, the MGM Smart Parking System provides a smarter and more efficient parking solution.

One of the major objectives of the system is to reduce traffic congestion inside parking areas. In traditional parking systems, drivers move around searching for empty parking slots, which creates unnecessary traffic and confusion. The MGM Smart Parking System helps users quickly identify available parking spaces and reduces unnecessary vehicle movement. Another important objective is to save time for users. Finding parking spaces manually can take several minutes, especially during busy hours. The system provides real-time parking information so that users can locate parking slots quickly and easily. The system is also designed to reduce fuel consumption. Vehicles moving continuously in search of parking spaces consume extra fuel and increase transportation costs. By providing organized parking management, the system helps minimize fuel wastage. Automation is another major objective of the MGM Smart Parking System. Traditional parking management depends heavily on manual work, which increases the chances of human errors. The smart parking system automates parking activities such as vehicle entry, exit monitoring, parking slot allocation, and record maintenance. Maintaining accurate records is also an important objective of the system. All parking-related information, including vehicle details and parking status, is stored securely in a digital database. This improves data accuracy and simplifies parking management.

The system also focuses on improving security within parking areas. Digital monitoring helps administrators track vehicle movements and maintain secure parking operations. Unauthorized vehicle access and parking-related security issues can be controlled more effectively.

Another objective is to provide real-time parking information. Users can instantly check parking slot availability, which improves convenience and reduces waiting time.

Finally, the MGM Smart Parking System supports smart infrastructure development. The system contributes to the concept of smart campuses and smart cities by using modern technologies to improve transportation and parking management.

1.6 Features of MGM Smart Parking System

The MGM Smart Parking System includes several advanced features that help improve parking management and user experience. These features make the system more efficient, reliable, and user-friendly compared to traditional parking systems.

One of the most important features of the system is real-time parking availability. The system continuously updates parking slot status and displays available parking spaces instantly. This helps users locate parking spaces quickly without unnecessary searching. The system also provides a user-friendly interface. Both users and administrators can operate the system easily because of its simple design and organized structure. The interface helps users access parking information quickly and efficiently. Vehicle registration is another important feature of the system. Users can register their vehicle details digitally, including vehicle number, owner name, and contact information. This information is stored securely in the database for future reference.

The MGM Smart Parking System also supports automated entry and exit monitoring. Vehicle movements are tracked automatically, and entry and exit times are recorded in the system database. This improves parking management accuracy and reduces manual work. Database management is an essential feature of the system. All parking-related data, including parking slot information, user details, and vehicle records, are stored digitally in a secure database. This helps administrators maintain accurate parking records. Parking slot management is another important feature. The system efficiently manages available and occupied parking slots and updates parking status in real time. Security monitoring also plays an important role in the system. Administrators can monitor parking activities and vehicle movements digitally, which improves parking security and reduces the chances of unauthorized access.

The system can also generate parking reports and records. Administrators can access parking statistics, parking history, and vehicle details whenever required. This helps improve parking management and decision-making.

1.7 Applications of MGM Smart Parking System

The MGM Smart Parking System can be implemented in many places where parking management is important. The system is suitable for both small and large parking areas because it improves parking efficiency and reduces management difficulties.

In colleges and universities, the system can be used to manage student, faculty, and visitor parking. Educational institutions often face parking congestion during peak hours, and the smart parking system helps organize parking activities efficiently. Hospitals can also benefit from the MGM Smart Parking System. Emergency vehicles, doctors, staff, and visitors require organized parking facilities. The system helps hospitals manage parking areas effectively and reduce congestion. Shopping malls and commercial centers can use the system to improve customer parking management. Customers can quickly find available parking spaces, which improves their shopping experience. Airports and railway stations also require efficient parking management because of high vehicle traffic. Smart parking systems help reduce congestion and improve vehicle movement in these transportation areas. Office buildings and corporate organizations can use the system to manage employee and visitor parking efficiently. The system helps improve parking organization and security. Residential apartments and housing societies can also implement smart parking systems to improve parking security and reduce parking conflicts among residents.

Smart city projects widely use smart parking systems because they support intelligent transportation management and improve urban infrastructure.

1.8 Technologies Used in MGM Smart Parking System

Several modern technologies are used in the development of the MGM Smart Parking System. These technologies help improve system performance, automation, and user experience.

Frontend technologies such as HTML, CSS, and JavaScript are used to design the user interface. HTML is used to create webpage structure, CSS is used for styling and layout design, and JavaScript is used to add interactivity and dynamic functionality to the system.

Backend technologies such as Python and Flask Framework are used to process user requests and manage system operations. Python is widely used because of its simplicity and efficiency, while Flask helps develop web applications easily. The MySQL database is used to store parking records, user information, vehicle details, parking slot data, and system reports. The database helps maintain accurate and secure parking information.

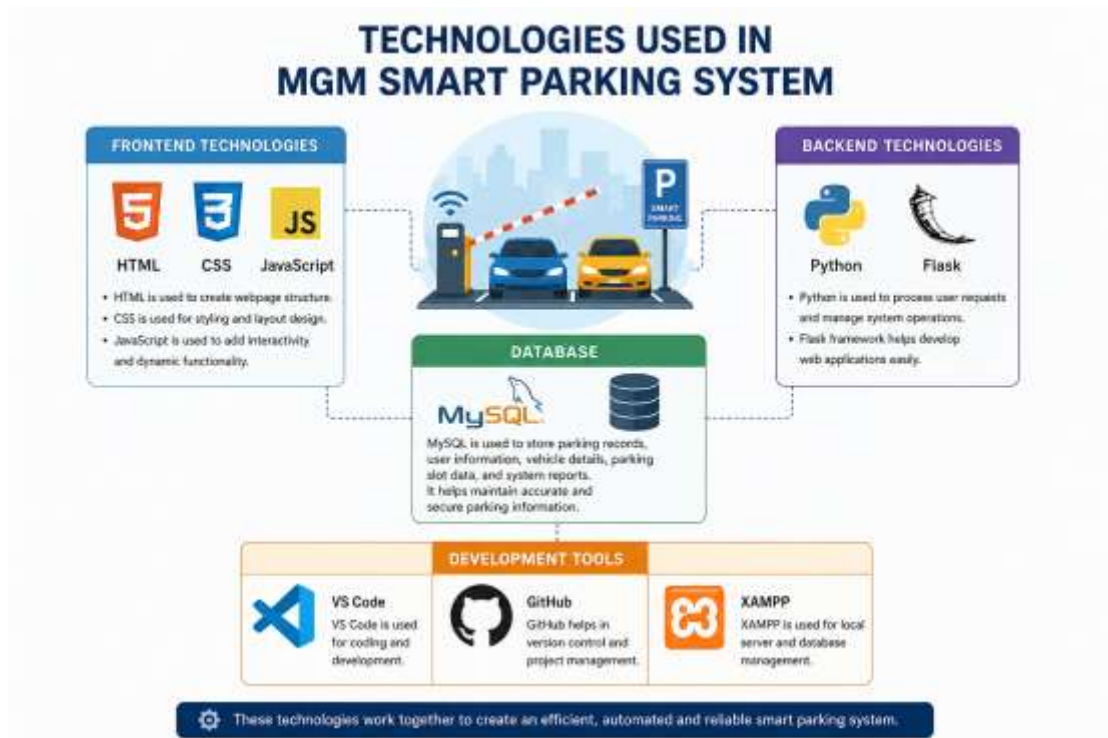


Fig. 1.1: Technologies Used

Development tools such as VS Code, GitHub, and XAMPP are also used during project development. VS Code is used for coding, GitHub helps manage project versions, and XAMPP supports local server and database management. These technologies work together to create an efficient and reliable smart parking system.

1.9 Future Scope of MGM Smart Parking System

The future scope of the MGM Smart Parking System is very wide because modern technologies are continuously improving transportation and parking management systems. As cities become more crowded and the number of vehicles increases every year, smart parking solutions will play an important role in creating organized and efficient transportation systems. The MGM Smart Parking System can be further enhanced by integrating advanced technologies that improve automation, security, accuracy, and user convenience.

One of the major future improvements of the system is the integration of Internet of Things (IoT) technology. IoT sensors can be installed in parking slots to automatically detect whether a parking space is occupied or empty. These sensors can send real-time information to the system, helping users quickly identify available parking spaces without wasting time. This will reduce traffic congestion and improve parking

efficiency. RFID technology can also be added to the system for automatic vehicle identification and secure access control. RFID tags attached to vehicles can help the system recognize authorized vehicles automatically during entry and exit. This reduces manual verification work and improves parking security.

Another important future enhancement is mobile application support. A dedicated mobile application can allow users to check parking availability, reserve parking slots, receive notifications, and manage parking activities directly from their smartphones. This feature will provide greater convenience and improve the user experience.

The system can also support online payment facilities in the future. Users may be able to make digital payments for parking charges through mobile wallets, UPI, debit cards, or online banking systems. This will reduce manual cash handling and make payment processing faster and easier. Artificial Intelligence (AI) can also be integrated into the system for intelligent parking management. AI algorithms can analyze parking data, monitor vehicle movement patterns, and predict parking availability during different time periods. This will help administrators manage parking areas more efficiently. Cloud computing technology can improve data storage and system accessibility. Cloud-based parking management systems can store large amounts of parking data securely and allow users to access parking information from different locations. Future smart parking systems may also include automatic number plate recognition, smart notifications, voice-based assistance, and electric vehicle charging support. These technologies will make the system more advanced, intelligent, and suitable for modern smart city infrastructure.

LITERATURE REVIEW

The literature review is an important section of the project report because it helps in understanding the existing parking management systems, technologies, methods, and research work related to smart parking solutions. It provides knowledge about how different parking systems work, their advantages, limitations, and the improvements required for better parking management.

Parking management has become a major issue in modern institutions, commercial areas, hospitals, shopping malls, and educational campuses due to the increasing number of vehicles. In many places, traditional parking systems are still used, where parking activities are managed manually. Drivers often spend a lot of time searching for parking spaces, which creates traffic congestion, fuel wastage, and inconvenience.

Many researchers and developers have worked on improving parking systems by introducing automated and digital parking management solutions. These systems focus on reducing manual work, improving parking efficiency, maintaining parking records, and providing better security. The MGM Smart Parking System is developed after studying different parking management approaches and understanding the problems faced in existing systems.

2.1 Study of Traditional Parking Systems

Traditional parking systems are mostly manual and depend heavily on human monitoring. In such systems, security guards or parking staff manage vehicle entry and exit activities manually. Parking records are maintained using notebooks, registers, or basic computer systems.

Researchers observed that traditional parking systems create several operational difficulties. Drivers often face confusion while searching for parking spaces because there is no proper indication of parking availability. During peak hours, parking areas become overcrowded, which increases traffic congestion and delays. Another major issue identified in traditional systems is time wastage. Drivers spend several minutes searching for parking slots, especially in large parking areas. This unnecessary movement of vehicles increases fuel consumption and creates environmental pollution.



Fig. 2.1: Traditional Parking Problem

Manual parking systems also create problems in record maintenance. Parking data may be recorded incorrectly or lost due to human errors. Administrators find it difficult to monitor parking activities efficiently because of the lack of proper digital management systems. Security is another challenge in traditional parking systems. Since parking monitoring is mostly manual, unauthorized parking activities and vehicle-related security issues may occur. Researchers concluded that traditional parking methods are not suitable for handling large parking areas efficiently.

2.2 Review of Existing Digital Parking Systems

To solve the problems found in manual parking systems, many researchers introduced digital parking management systems. These systems use web applications, database management systems, automation techniques, and software-based monitoring to improve parking operations. Several studies focused on web-based parking systems where users can check parking information using a web portal. These systems allow administrators to maintain parking records digitally and monitor parking activities more efficiently.

Researchers found that digital parking systems improve parking management by reducing paperwork and manual operations. Parking slot information can be updated

automatically, and parking records can be stored securely in databases. Some parking management systems include user login features that help maintain secure access to parking information. Vehicle details, parking history, and parking status can be managed digitally, which improves accuracy and reduces human errors.

Many existing systems also include automated parking slot allocation methods. These systems help users identify available parking spaces quickly, reducing congestion and improving parking efficiency. Although digital parking systems provide several advantages, researchers observed that some systems are difficult to use because of complex interfaces and poor user experience. Certain systems also require expensive hardware and maintenance costs, making them less suitable for smaller organizations.

2.3 Research on Web-Based Parking Management Systems

Web-based parking systems are widely used because they are easy to access and manage. Researchers developed several web-based parking applications to improve parking organization and reduce manual work.

In these systems, users can access parking information through web browsers. Administrators can monitor parking slots, maintain parking records, and manage vehicle entry and exit activities digitally. Studies showed that web-based systems improve parking management efficiency because all parking information is stored in a centralized database. This allows easy access to parking records and simplifies parking administration.

Researchers also found that web-based parking systems reduce paperwork and improve communication between users and administrators. Parking data can be updated in real time, helping users identify available parking slots more quickly. Web-based systems are also cost-effective compared to advanced hardware-based parking systems because they mainly depend on software applications and database management.

The MGM Smart Parking System is also designed as a web-based parking management system because web technologies provide flexibility, easy access, and efficient parking control.

2.4 Database Management in Parking Systems

Database management plays an important role in smart parking systems. Researchers studied different database systems used for storing parking information securely and efficiently. Most parking systems use relational databases to store vehicle details, parking slot information, entry and exit records, user details, and parking reports. Databases help maintain accurate records and simplify parking management.

Researchers found that digital databases reduce the chances of data loss and improve information accessibility. Administrators can retrieve parking information quickly whenever required. Database systems also help generate parking reports and parking statistics automatically. This improves parking monitoring and helps administrators analyze parking usage patterns.

The MGM Smart Parking System uses database management to maintain organized parking records and improve system reliability.

2.5 Problems Identified in Existing Systems

During the literature review, several limitations were identified in existing parking systems. Some systems require expensive hardware and maintenance costs, which may not be affordable for small institutions and organizations. Certain systems are difficult for users to operate because of poor interface design and complicated navigation.

Researchers also observed that some parking systems do not provide proper real-time parking updates, which reduces parking efficiency. Manual verification processes in some systems create delays during vehicle entry and exit operations.

Problems	Description
Traffic Congestion	Vehicles move continuously searching for parking
Time Wastage	Drivers spend extra time finding parking spaces
Fuel Consumption	More fuel usage during parking search
Manual Errors	Incorrect record maintenance
Poor Security	Lack of proper monitoring
Difficult Management	Large parking areas are difficult to manage

Table 2.1: Problems in Traditional Parking Systems

Lack of proper security monitoring is another issue found in several parking systems. Unauthorized parking activities may occur if monitoring systems are weak. These limitations highlighted the need for a more user-friendly, reliable, and cost-effective parking management solution like the MGM Smart Parking System.

2.6 Proposed MGM Smart Parking System

After studying different parking management systems and analyzing the problems found in traditional parking methods, the MGM Smart Parking System was developed as a modern and efficient parking management solution. The main purpose of the proposed system is to simplify parking operations, reduce manual work, and improve overall parking efficiency within the organization or campus area.

In many institutions and public places, parking management becomes difficult because of the increasing number of vehicles and limited parking spaces. Traditional parking systems depend mainly on manual monitoring, where security guards or parking staff maintain vehicle records and guide drivers to available parking spaces. This process consumes more time and often creates confusion, traffic congestion, and parking mismanagement. To overcome these problems, the MGM Smart Parking System introduces a digital and automated approach to parking management. The proposed system uses web technologies, database management systems, and automation concepts to provide a reliable and user-friendly parking solution. The system allows users to access parking information digitally and helps administrators monitor parking activities more effectively. By replacing manual operations with computerized management, the system improves accuracy and reduces human errors in parking record maintenance. One of the important features of the proposed system is parking slot availability management. Users can check available parking spaces through the system before parking their vehicles. This reduces unnecessary movement inside parking areas and helps save time and fuel. The system also updates parking status regularly, which improves parking organization and reduces traffic congestion.

The MGM Smart Parking System also includes vehicle registration and parking record management features. Users can register their vehicle details in the system, and all parking-related information is stored securely in the database. Administrators can easily access parking records, monitor vehicle entry and exit activities, and manage parking operations efficiently. Another important aspect of the proposed system is security

improvement. Since all parking information is maintained digitally, administrators can monitor parking activities more effectively. Vehicle movement records can be tracked easily, which improves parking safety and reduces the chances of unauthorized parking activities. The system is also designed with a simple and user-friendly interface so that users and administrators can operate it without difficulty. The web-based design allows easy access to parking information and improves communication between users and parking management authorities.

The MGM Smart Parking System also helps reduce paperwork and manual documentation. Traditional parking systems require continuous manual record maintenance, which increases workload and the possibility of errors. The proposed system automates parking record management and improves data accuracy.

SYSTEM DESIGN

System design is an important stage in the development of the MGM Smart Parking System because it explains how the entire parking management system is planned, organized, and developed to work smoothly and efficiently. A well-designed system helps in understanding the overall structure of the project, the flow of information, the connection between different modules, and the management of parking operations. It also helps developers and users understand how different parts of the system communicate with each other to perform parking-related activities successfully.

The MGM Smart Parking System is designed to provide a simple, reliable, and user-friendly parking management solution. In many places, traditional parking systems still depend on manual work, which creates problems such as traffic congestion, time wastage, fuel consumption, poor record maintenance, and parking confusion. To overcome these issues, the MGM Smart Parking System introduces a digital approach that automates parking operations and improves overall parking management. The system is developed in such a way that users can easily check parking slot availability, register vehicle details, monitor parking status, and manage parking activities digitally. Administrators can also monitor parking records, track vehicle movement, and generate reports more efficiently. By replacing manual parking methods with an automated system, the project helps improve accuracy, reduce paperwork, and simplify parking operations. The design of the MGM Smart Parking System mainly focuses on simplicity, security, efficiency, and easy accessibility. The system is created with a user-friendly interface so that users and administrators can operate it without difficulty. The organized design of the system helps reduce unnecessary vehicle movement inside parking areas, which ultimately reduces traffic congestion and saves time.

The MGM Smart Parking System is developed using web technologies and database management systems. All parking-related information is stored digitally in the database, which improves data accuracy and reduces the possibility of human errors. Digital data management also helps administrators retrieve parking information quickly whenever required.

Another important aspect of the system design is proper monitoring and management of parking activities. The system continuously updates parking information and

maintains accurate records of vehicle entry, exit, and parking slot status. This improves parking organization and provides better control over parking operations.

Overall, the system design of the MGM Smart Parking System provides a strong foundation for building an efficient, secure, and reliable parking management solution. It helps improve parking efficiency, reduces manual workload, enhances data management, and supports smart digital infrastructure development.

3.1 Overview of System Design

The MGM Smart Parking System is designed using a web-based architecture that allows both users and administrators to access the system through an online platform. The system performs all parking-related activities digitally, which helps improve parking management and reduce manual work. All parking information, including vehicle details, parking slot status, user records, and parking history, is stored securely in a database for easy access and management.

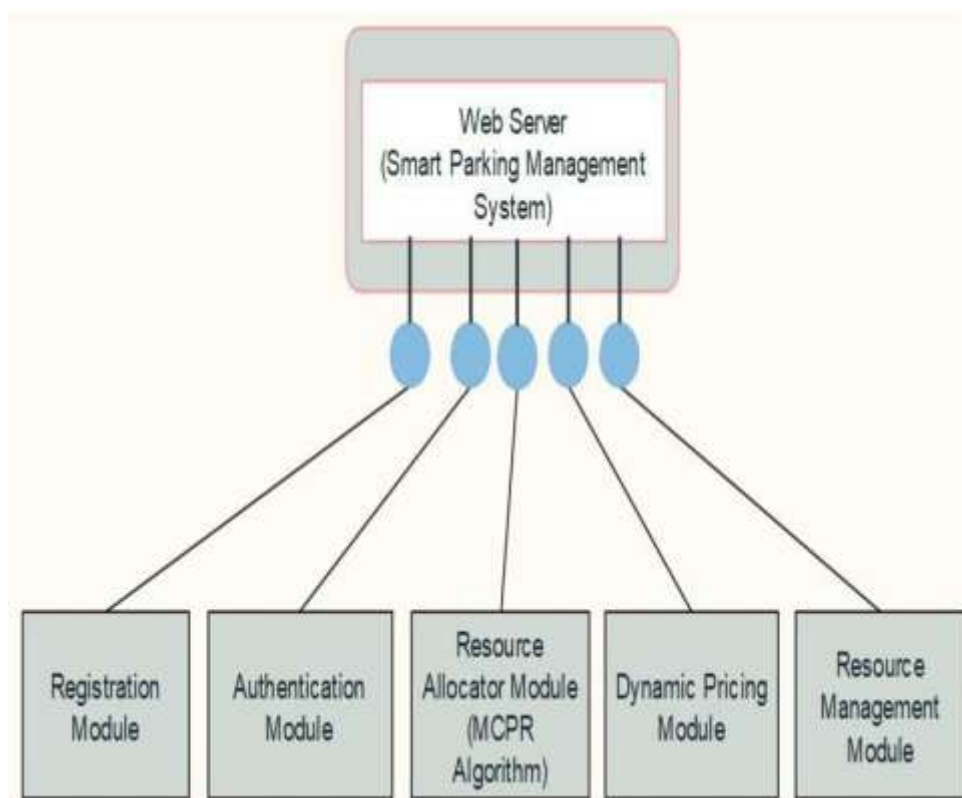


Fig 3.1: Overview of Parking System

The overall system is divided into different modules, and each module is responsible for performing a specific task related to parking management. This modular design

helps make the system more organized, efficient, and easy to manage. All modules work together smoothly to ensure proper communication and effective parking operations. The MGM Smart Parking System mainly includes modules such as user management, vehicle registration, parking slot management, parking monitoring, database management, and report generation. The user management module controls user access and authentication, while the vehicle registration module stores vehicle-related information. The parking slot management module helps manage available and occupied parking spaces efficiently. Parking monitoring helps track vehicle movement and parking activities in real time. The database module securely stores all parking records, and the report generation module helps administrators generate parking reports and parking statistics whenever required.

The system is designed with a user-friendly interface so that users can easily access parking information without confusion. Administrators can also monitor parking operations, manage records, and control parking activities efficiently through the system dashboard. The organized structure of the system helps improve parking efficiency, reduce paperwork, and simplify parking management operations. Another important feature of the system design is proper communication between different layers of the system. The user interface, application processing section, and database are connected properly to ensure smooth data transfer and efficient system performance. Whenever users perform any activity, such as checking parking availability or registering vehicle details, the information is processed by the application layer and stored securely in the database. The web-based design of the MGM Smart Parking System also improves accessibility because users and administrators can access the system from different devices connected to the internet. This makes the parking management process more flexible, convenient, and efficient.

3.2 System Architecture

The architecture of the MGM Smart Parking System explains how different components of the system interact with each other. The system architecture is designed to provide better performance, easy maintenance, and efficient parking management.

The system mainly consists of three important layers:

3.2.1 Presentation Layer: The presentation layer represents the user interface of the system. This layer allows users and administrators to interact with the system through webpages. The interface is developed using HTML, CSS, and JavaScript. The presentation layer helps users perform activities such as checking parking availability, entering vehicle details, viewing parking status, and accessing parking information. The interface is designed in a simple and understandable manner so that users can operate the system easily.

3.2.2 Application Layer: The application layer works as the main processing section of the system. It handles user requests and performs different parking management operations. This layer is developed using Python and Flask Framework. The application layer processes parking data, verifies user login details, manages parking slot allocation, updates parking status, and controls vehicle entry and exit operations. It acts as a connection between the user interface and the database system. This layer helps improve automation and ensures smooth functioning of the parking management process.

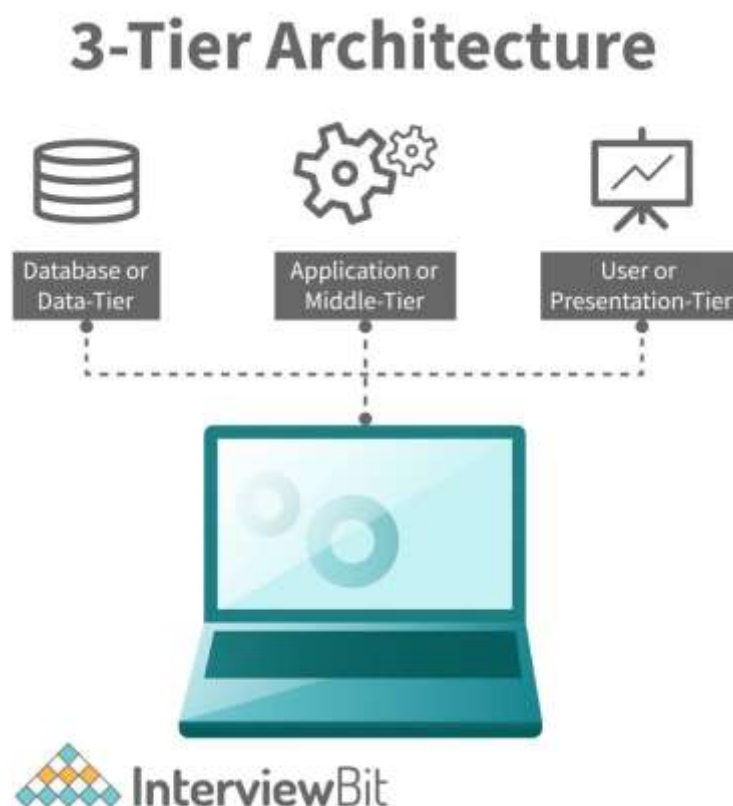


Fig 3.2 System Architecture

3.2.3 Database Layer: The database layer stores all parking-related information securely. The MGM Smart Parking System uses MySQL database for storing user details, vehicle information, parking slot records, and parking reports. The database helps maintain accurate records and allows administrators to retrieve parking information whenever required. It also improves data organization and simplifies parking management activities.

The database layer ensures data security, reliability, and efficient storage management.

3.3 Modules of MGM Smart Parking System

The MGM Smart Parking System is divided into several important modules to improve system organization and simplify parking management activities. Each module performs a specific function and helps the overall system operate smoothly and efficiently. These modules work together to reduce manual work, improve parking monitoring, maintain accurate records, and provide better parking management for users and administrators.

3.3.1 Login Module

The login module is responsible for providing secure access to the system. Users and administrators must enter valid login credentials such as username and password to access system features. This module helps protect parking information from unauthorized access and improves overall system security. The login module also helps identify different users and administrators separately. Only authorized persons can manage parking operations and access parking records. This improves system reliability and maintains proper security control within the parking management system.

Main Functions:

- User authentication
- Password verification
- Secure system access
- Prevention of unauthorized access
- Administrator login management

3.3.2 Vehicle Registration Module

The vehicle registration module allows users to enter and store vehicle-related information in the system. Details such as vehicle number, owner name, contact number, and vehicle type are stored securely in the database. This module helps maintain proper vehicle records and simplifies parking management activities. Administrators can easily retrieve vehicle information whenever required. Digital vehicle registration also reduces paperwork and minimizes manual errors.

Main Functions:

- Vehicle detail registration
- Owner information storage
- Vehicle number management
- Digital record maintenance
- Secure data storage

3.3.3 Parking Slot Management Module

The parking slot management module helps manage available and occupied parking spaces efficiently. This module continuously updates parking slot status and displays parking availability information to users. When a vehicle enters the parking area, the system allocates an available parking slot automatically. Once the vehicle exits, the parking slot becomes available again. This process improves parking organization and reduces confusion while searching for parking spaces. The module also helps reduce unnecessary vehicle movement inside parking areas, which minimizes traffic congestion and fuel wastage.

Main Functions:

- Parking slot allocation
- Parking availability monitoring
- Occupied and vacant slot management
- Real-time parking status updates
- Reduction of parking confusion

3.3.4 Entry and Exit Monitoring Module

The entry and exit monitoring module records vehicle movement details automatically. Whenever a vehicle enters or exits the parking area, the system stores entry time, exit time, and parking history in the database. This module improves parking monitoring and helps administrators track vehicle activities efficiently. It also increases parking security because vehicle movement records can be verified whenever required. Accurate entry and exit monitoring helps maintain proper parking records and supports better parking management.

Main Functions:

- Vehicle entry recording
- Vehicle exit recording
- Entry and exit time management
- Parking history maintenance
- Vehicle movement tracking

3.3.5 Database Management Module

The database management module stores and manages all parking-related information securely. The system uses a database to maintain user records, vehicle information, parking slot details, parking history, and parking reports. The database helps organize information properly and allows quick retrieval of parking data whenever needed. It improves data accuracy and reduces the possibility of information loss. Digital data management also simplifies parking administration and supports efficient system performance.

Main Functions:

- Secure data storage
- Parking record management
- User information maintenance
- Vehicle data management

3.5 Output Design

Output design refers to the information displayed by the MGM Smart Parking System after processing user requests and parking operations. A proper output design helps

users and administrators understand parking information clearly and improves communication between the system and users. The MGM Smart Parking System provides different types of outputs such as parking slot availability, parking status updates, vehicle entry and exit records, parking reports, notifications, and parking history details. These outputs are displayed in a clear and organized format so that users can easily access parking information without confusion. One of the important outputs of the system is parking slot availability information. Users can quickly view available and occupied parking slots through the system interface. This helps reduce unnecessary searching for parking spaces and improves parking efficiency.

The system also displays vehicle entry and exit details. Whenever a vehicle enters or exits the parking area, the information is updated automatically and shown in the system records. This helps administrators monitor parking activities properly. Parking reports and parking statistics are another important part of output design. Administrators can generate reports related to parking usage, vehicle records, parking history, and parking activities. These reports help improve parking management and decision-making.

The MGM Smart Parking System also provides notifications and status updates to users whenever required. Important information such as successful vehicle registration, parking slot allocation, and parking updates can be displayed through the system interface.

The outputs are designed in a simple, clear, and user-friendly format so that both users and administrators can understand the information easily. Proper output design improves system usability, reduces confusion, and supports efficient parking management.

IMPLEMENTATION

The implementation phase is one of the most important stages in the development of the MGM Smart Parking System. In this phase, the planned system design is converted into a real working application using web technologies and database management systems. The implementation process focuses on developing a simple, reliable, and efficient parking management solution that can perform parking operations smoothly and accurately.

The MGM Smart Parking System is developed to reduce the problems found in traditional parking systems such as traffic congestion, manual record maintenance, parking confusion, and time wastage. The system provides a digital platform where users and administrators can manage parking activities easily. Through the implemented system, users can register vehicle details, check parking availability, monitor parking status, and maintain parking records digitally. The implementation of the system improves parking management by reducing manual work and increasing operational efficiency. It also helps administrators monitor parking activities properly and maintain organized parking records. The system is designed with a user-friendly interface so that users can access parking information easily without technical difficulty.

The MGM Smart Parking System is implemented using frontend technologies such as HTML, CSS, and JavaScript for interface design and user interaction. Backend processing is handled using Python and Flask Framework, while MySQL database is used to store parking-related information securely. These technologies work together to provide smooth system performance and reliable parking management.

4.1 Technologies Used

The MGM Smart Parking System is implemented using modern web development technologies and database management tools. These technologies help improve system performance, automation, security, and user experience.

4.1.1 Frontend Technologies:

HTML: HTML is used to create the basic structure of webpages and forms used in the parking system.

CSS: CSS is used for webpage styling, layout design, colors, buttons, and improving the visual appearance of the system.

JavaScript: JavaScript is used to add dynamic functionality and improve user interaction with the system.

These frontend technologies help create a simple and user-friendly interface for parking management.



Fig 4.1: Technology Stack

4.1.2 Backend Technologies

Python: Python is used for backend programming and system processing because of its simplicity and efficiency.

Flask Framework: Flask Framework is used for developing the web application and connecting the user interface with the database.

The backend technologies help process user requests, manage parking operations, and control parking activities efficiently.

4.4.3 Database

MySQL: MySQL database is used for storing user details, vehicle records, parking slot information, parking history, and parking reports securely.

The database helps organize parking information properly and supports accurate parking management.

4.2 System Implementation

The MGM Smart Parking System is implemented as a web-based parking management application. Users and administrators can access the system through webpages and manage parking activities digitally.



Fig 4.2: Admin Login

The login system is implemented to provide secure access using usernames and passwords. Only authorized users can access the system, which improves parking security and protects parking records from unauthorized access.

The vehicle registration section allows users to enter vehicle details such as vehicle number, owner name, and contact information. These details are stored securely in the database for future parking management.



Fig 4.3: Sign Up Page

The parking slot management section continuously updates available and occupied parking spaces. When a vehicle enters the parking area, the system allocates an available parking slot automatically. After vehicle exit, the parking slot becomes available again. This process improves parking organization and reduces confusion while searching for parking spaces.

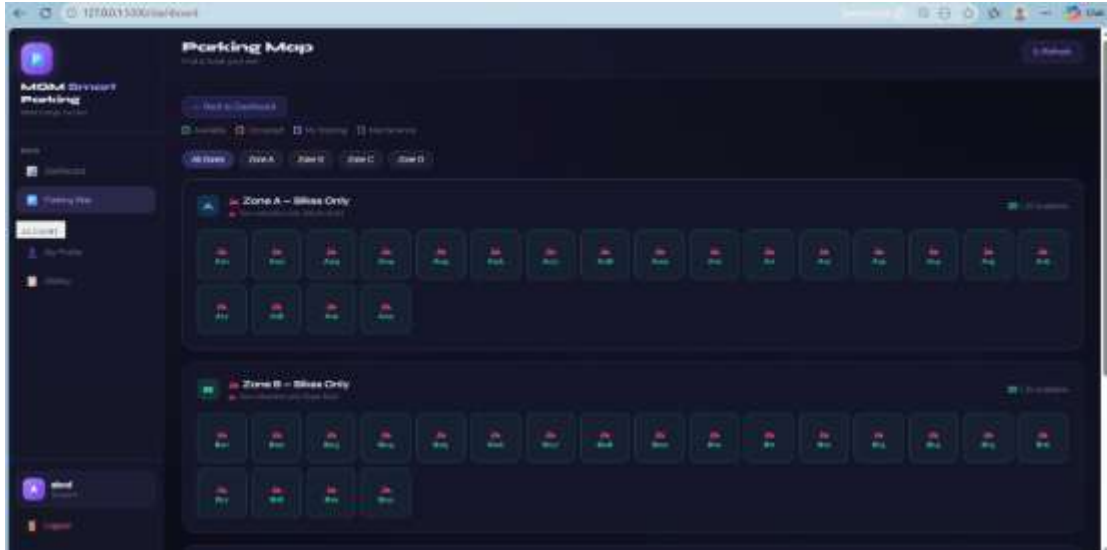


Fig 4.4: Parking Slots

The entry and exit monitoring system records vehicle movement details automatically. Entry time, exit time, and parking history are stored in the database, which improves parking monitoring and security management.

The database management system stores all parking-related information in a structured format. Parking reports and parking history can also be generated whenever required. This helps administrators monitor parking activities efficiently and maintain proper parking records.

4.3 Advantages of Implementation

The implementation of the MGM Smart Parking System provides several advantages compared to traditional parking systems.

- Improves parking management efficiency
- Reduces manual work and paperwork
- Maintains accurate parking records
- Improves parking security and monitoring

- Reduces traffic congestion inside parking areas
- Saves time for users and administrators
- Simplifies parking slot management
- Supports digital parking operations
- Reduces human errors in parking management
- Improves overall parking organization



Fig 4.: Benefits of Smart Parking System

The implemented system creates a better parking experience for users and simplifies parking administration activities.

TESTING AND EVALUATION

Testing and evaluation are very important stages in the development of the MGM Smart Parking System because they help ensure that the system works correctly, efficiently, and without major errors. The main purpose of testing is to verify whether all modules and functions of the parking system are operating properly according to the planned design. Evaluation helps measure the overall performance, reliability, security, and usability of the system.

The MGM Smart Parking System was tested carefully to make sure that users and administrators could use the system smoothly and efficiently. Different functions such as user login, vehicle registration, parking slot allocation, parking monitoring, report generation, and database management were tested during the development process. The testing process helped identify errors and improve system performance before final implementation.

The evaluation process also helped analyze whether the system successfully reduced manual parking operations, improved parking management, and provided accurate parking information. Proper testing and evaluation increased the reliability and efficiency of the system and ensured better parking management operations.

5.1 Objectives of Testing

The main objectives of testing the MGM Smart Parking System are to verify proper system functionality and improve the overall quality of the project. Testing helps ensure that the system performs parking operations correctly and provides accurate parking information.

The important objectives of testing include:

- To verify smooth system performance
- To identify and remove system errors
- To improve parking management efficiency
- To verify secure user authentication
- To check proper parking slot allocation
- To maintain accurate parking records

- To improve database reliability
- To ensure proper communication between modules
- To provide a user-friendly parking system
- To improve overall system security and stability

5.2 Types of Testing Performed

Different testing methods were used during the development of the MGM Smart Parking System to ensure smooth and reliable system performance.

5.2.1 Unit Testing:

Unit testing was performed to test individual modules separately. Each module such as login, vehicle registration, parking slot management, and report generation was tested independently to verify correct functionality.

This testing helped identify errors at an early stage and improved module reliability.

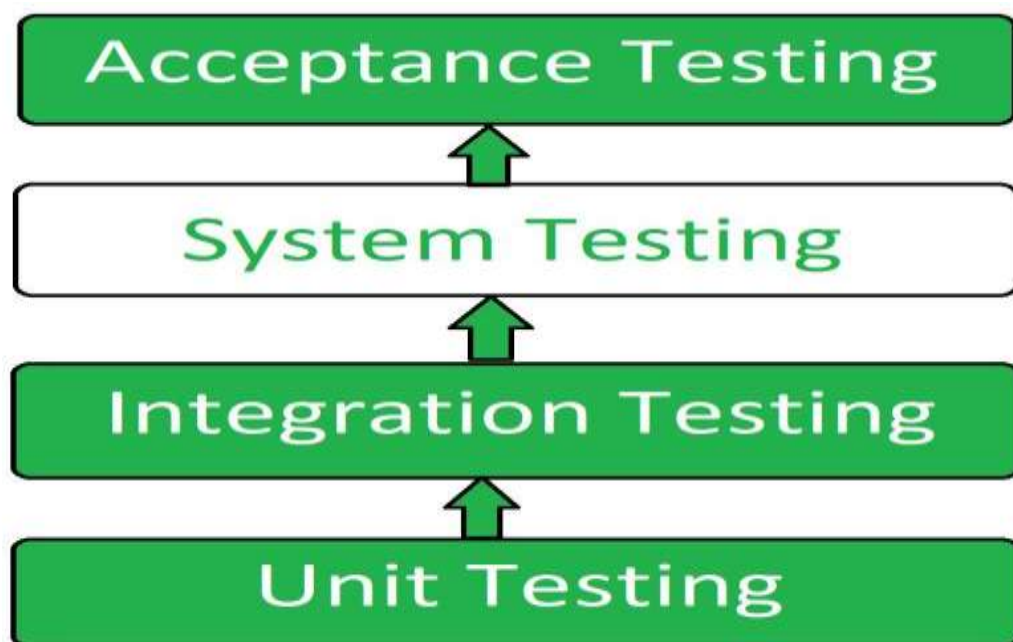


Fig 5.1: Testing Performance

5.2.2 Integration Testing:

Integration testing was used to verify communication between different modules of the system. This testing checked whether all modules worked together properly without any data transfer problems.

The testing ensured smooth interaction between the user interface, backend processing, and database management system.

5.2.3 System Testing:

System testing was performed on the complete MGM Smart Parking System after integrating all modules. This testing verified the overall functionality and performance of the parking management system.

The system was tested under different conditions to ensure smooth parking operations and reliable performance.

5.2.4 User Interface Testing

User interface testing focused on verifying webpage design, navigation, buttons, forms, and system usability. The testing ensured that users could easily operate the system without confusion.

The interface was tested on different devices and browsers to improve accessibility and user experience.

5.2.5 Database Testing

Database testing was performed to verify secure data storage and proper retrieval of parking records, vehicle information, and parking history.

This testing helped ensure accurate parking record management and improved database reliability.

5.3 Testing of System Modules

Module testing was performed to ensure that every component of the MGM Smart Parking System functioned correctly and met the expected requirements. Each module was tested separately to verify its performance, accuracy, and reliability before integrating it with the complete system. The testing process helped identify potential errors and ensured that all parking management operations worked smoothly without interruption.

The MGM Smart Parking System modules, including Login, Vehicle Registration, Parking Slot Management, Entry and Exit Monitoring, Database Management, and Report Generation, were tested individually to verify proper functionality.

5.3.1 Login Module Testing

The login module was tested using both valid and invalid login credentials. The system successfully allowed authorized users and rejected incorrect login attempts. The module also verified user authentication and prevented unauthorized access to parking information.

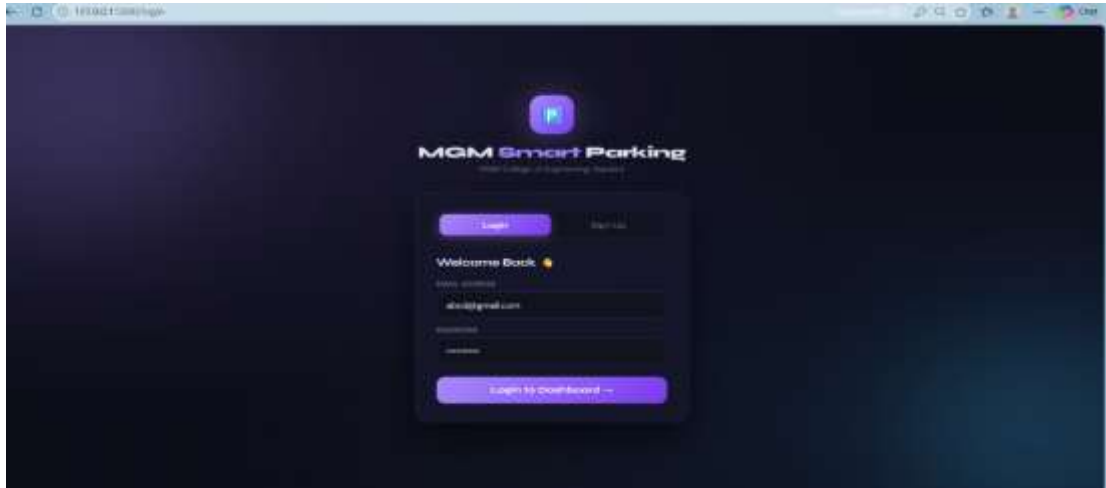


Fig 5.2: User Login

Result: The login system worked correctly and provided secure access control.

5.3.2 Vehicle Registration Module Testing

The vehicle registration module was tested by entering different vehicle details such as vehicle number, owner information, and contact details. The system stored all information correctly in the database and displayed proper confirmation messages.

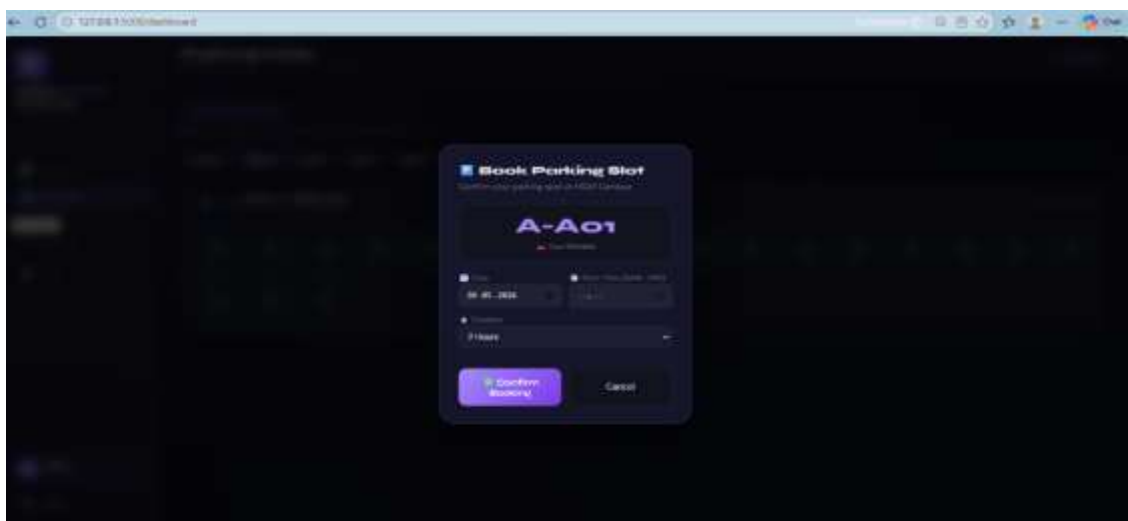


Fig 5.3: Parking Booking Slot

Result: Vehicle registration and data storage worked successfully.

5.3.3 Parking Slot Management Testing

The parking slot management module was tested to verify proper parking slot allocation and parking status updates. When vehicles entered or exited the parking area, the system updated parking slot availability automatically.

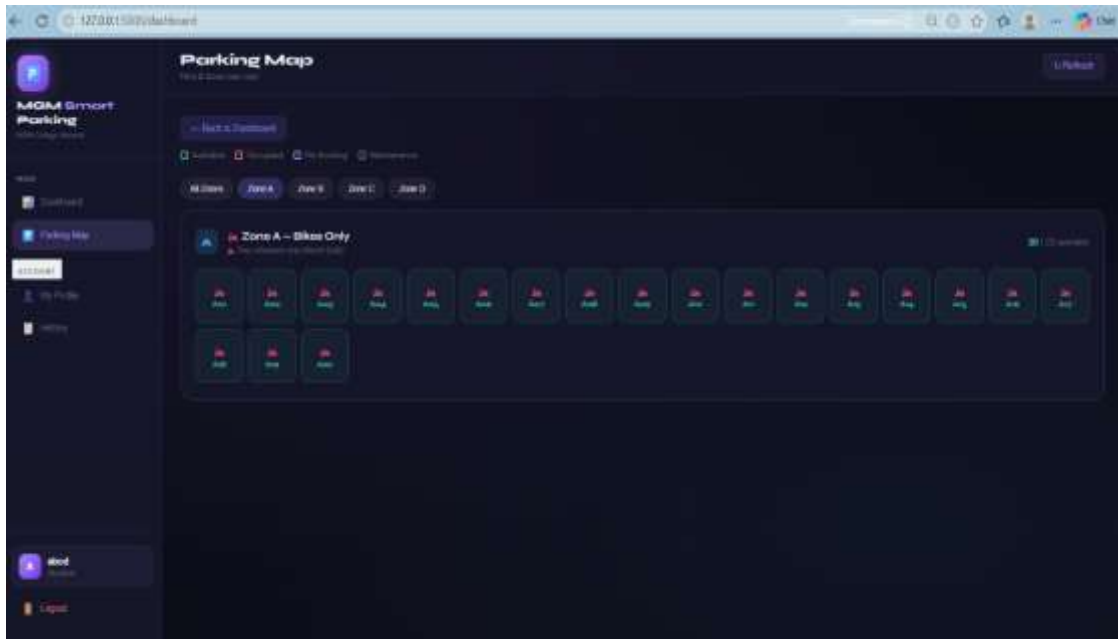


Fig 5.4: Parking Slots

Result: Parking slot management functioned accurately and efficiently.

5.3.4 Entry and Exit Monitoring Testing

The entry and exit monitoring module was tested by recording vehicle movement details during parking operations. The system maintained entry time, exit time, and parking history correctly in the database.

Result: Vehicle movement tracking worked properly and improved parking monitoring.

5.3.5 Database Management Testing

The database module was tested for secure data storage and retrieval of parking information. The system stored parking records, vehicle details, and parking reports accurately without data loss.

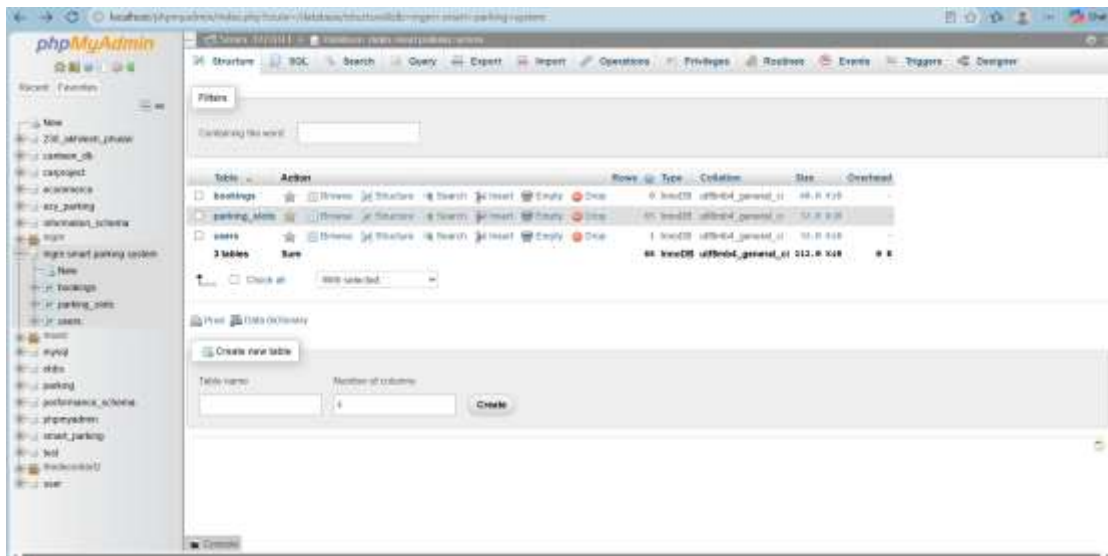


Fig 5.5: Database Connection

Result: Database management worked efficiently and maintained reliable parking records.

5.4 Evaluation of System Performance

The evaluation process helped measure the effectiveness and reliability of the MGM Smart Parking System. After testing all modules, the system showed stable and efficient performance during parking operations. The implemented system successfully reduced manual parking work and improved parking organization. Parking slot management became faster and more accurate compared to traditional parking systems.

Users were able to access parking information easily through the web-based interface, while administrators could monitor parking activities efficiently using digital parking records and reports.

The system also improved parking security by maintaining vehicle entry and exit records digitally. Accurate parking information and real-time parking updates helped reduce parking confusion and unnecessary vehicle movement.

5.5 Advantages Observed During Testing

Several advantages were observed during the testing and evaluation process of the MGM Smart Parking System.

- Improved parking management efficiency

- Accurate parking slot monitoring
- Faster parking information updates
- Reduced manual errors in parking operations
- Improved parking security and monitoring
- Easy access to parking records
- Organized parking management process
- Reliable database performance
- Better user experience
- Reduced paperwork and manual documentation

These advantages show that the system can provide effective parking management for institutions and organizations.

5.6 Limitations Observed During Testing

Although the system performed successfully, some limitations were identified during testing.

- The system depends on internet connectivity for web access
- Large parking areas may require additional system resources
- System performance may slow if database size increases significantly
- Continuous maintenance is required for smooth operation

These limitations can be improved in future system upgrades and enhancements.

Although the MGM Smart Parking System performed successfully during testing and evaluation, some limitations were identified while operating the system. These limitations may affect system performance under certain conditions and can be improved in future versions of the project.

One of the major limitations observed during testing is the dependency on internet connectivity. Since the MGM Smart Parking System is a web-based application, users and administrators require a stable internet connection to access parking information and perform parking operations. If the internet connection becomes slow or unavailable, some system functions may not operate properly. This may affect parking monitoring and reduce system accessibility.

Another limitation identified during testing is related to large parking areas and increasing numbers of users. The current system is suitable for small and medium-sized parking management environments, but very large parking areas may require additional system resources and server capacity for smooth performance. As the number of vehicles and users increases, the system may require better hardware and optimized database management to maintain efficiency. During testing, it was also observed that system security depends on proper user authentication and database protection. Although the current system provides secure login functionality, additional security measures may be required in future implementations to protect parking information from cyber threats and unauthorized access.

CONCLUSION

The MGM Smart Parking System was developed to provide a simple, efficient, and user-friendly solution for modern parking management problems. The system helps reduce manual parking operations by introducing a digital platform for vehicle registration, parking slot management, parking monitoring, and record maintenance. Through the use of web technologies and database management, the system improves parking organization and simplifies parking activities for both users and administrators.

The implemented system successfully reduces traffic congestion, parking confusion, and time wastage that commonly occur in traditional parking systems. The system maintains accurate parking records, improves parking security, and provides organized parking management through automated processes. Features such as secure login, parking slot updates, vehicle entry and exit monitoring, and report generation help improve overall system efficiency and reliability.

The MGM Smart Parking System provides an effective and reliable parking management solution suitable for institutions, campuses, offices, and organizations. The project demonstrates how digital technologies can improve parking operations and support smart infrastructure development. In the future, the system can be further enhanced with advanced features such as mobile application support, online parking reservation, cloud-based management, and automatic vehicle identification to improve system performance and user convenience.

REFERENCES

- [1] S. Sharma and R. Kumar, “Web-based smart parking management system,” *IEEE Journal of Smart Computing*, vol. 8, no. 3, June, 2021, pp. 45–52.
- [2] R. S. Pressman and B. Maxim, *Software Engineering: A Practitioner’s Approach*, 9th ed., New York: McGraw-Hill Education, 2019, pp. 1–900.
- [3] A. Jain and P. Verma, “Automated parking slot allocation using web technologies,” in *Proceedings of the IEEE International Conference on Smart Systems*, 2022, IEEE CODE 45872, pp. 110–116.
- [4] N. Patel and D. Shah, “Digital parking management for educational institutions,” *IEEE Journal of Advanced Computer Applications*, vol. 11, no. 2, March, 2020, pp. 33–40.
- [5] I. Sommerville, *Software Engineering*, 10th ed., London: Pearson Education, 2016, pp. 1–800.
- [6] M. Fowler, *UML Distilled: A Brief Guide to the Standard Object Modeling Language*, 3rd ed., Boston: Addison-Wesley, 2004, pp. 1–200.
- [7] T. Connolly and C. Begg, *Database Systems: A Practical Approach to Design, Implementation, and Management*, 6th ed., London: Pearson Education, 2015, pp. 1–1200.
- [8] K. Mehta and S. Joshi, “Efficient vehicle monitoring system using web applications,” in *Proceedings of the IEEE International Conference on Modern Web Technologies*, 2021, IEEE CODE 67421, pp. 89–95.
- [9] D. Flanagan, “Interactive web application development using JavaScript,” *IEEE Journal of Internet Technology*, vol. 9, no. 4, August, 2020, pp. 65–72.